HOMEWORK 2.6 · UNIT 2 · NC.8.EE.3 · NC.8.EE.1

# Scientific Notation (Intro)

Targets: EE.3-2, EE.3-1, EE.3-7, EE.1-3, EE.3-4, EE.1-5, EE.3-9, EE.3-8

Name _____ Date _____ Block _____

**Remember: Scientific notation** = $a \times 10^n$, where $1 \le a < 10$. Big number → **positive** exponent: $45{,}000 = 4.5 \times 10^4$. Small decimal → **negative** exponent: $0.0062 = 6.2 \times 10^{-3}$. The exponent = how many places the decimal point hops.

**SCORE 3.0** Approaching · the basics

Targets: EE.3-2, EE.3-1, EE.3-7, EE.1-3

**1** Write in standard form. $5.6 \times 10^6$ EE.3-2

**2** Write in scientific notation. $0.0091$ EE.3-1

**3** Write in scientific notation. $470{,}000{,}000$ EE.3-1

**4** Is $12.5 \times 10^6$ in scientific notation? EE.3-1,EE.3-7

Yes No

Why? _____

**5** $10^{-2} = -100$ REVIEW 2.3 EE.1-3

True False

**SCORE 4.0** Proficient · on target

Targets: EE.3-4, EE.3-7, EE.1-5

**6** The Sun is about $1 \times 10^9$ m wide. Earth is about $1 \times 10^7$ m wide. How many times as wide is the Sun? EE.3-4

(A) 100 times
(B) 2 times
(C) 10 times
(D) 1,000 times

**7** **Ty** made a mistake. Circle it. Explain it. Fix it. EE.3-7

TY'S WORK
$45{,}000 = 45 \times 10^3$

**8** Simplify. $(2x^3)^2 \cdot x^{-4}$ REVIEW 2.4 EE.1-5

**SCORE 5.0** Mastery · go deeper

Targets: EE.3-9, EE.3-8

**9** Order from least to greatest: $A = 3 \times 10^{-4}$, $B = 5 \times 10^{-6}$, $C = 2 \times 10^{-3}$. Explain how the exponents help. EE.3-9

The order is _____ . I looked at the exponents first because _____ .

**10** A human hair is about 0.00008 m wide. Write this in scientific notation. Why is scientific notation helpful here? EE.3-8

In scientific notation it is _____ . This helps because _____ .